BASIC ELECTRICAL PRINCIPLES

CHAPTER 1

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LEARNING OBJECTIVES

A student can understand the following in this Chapter

1. Basics of electricity
2. Electrical parameters and its definitions
3. Ohm's law and its verification
4. Kirchhoff's law
5. Working of Resistors
6. Working of Capacitors

1.1 INTRODUCTION TO ELECTRICITY

Electricity is the integral part of human lives like water, space, land, fire and air. Today's technological era, all the activities and functions of the human beings are pervaded/controlled/facilitated by electrical and electronic gadgets. For example, a person awakening from sleep and during sleep, the environment, work, routine physiological activities, etc., have been pervaded by electrical or electronics equipment like fan, computer, cell phone, etc. Presently, all fields of science, engineering, technology, humanities and arts mostly rely on electrical and electronic gadgets for their sustenance. Further, our body activities are initiated and controlled by movement of ions across the cell wall, which is called as bioelectricity. Currently, in medical diagnosis, the measurement of

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